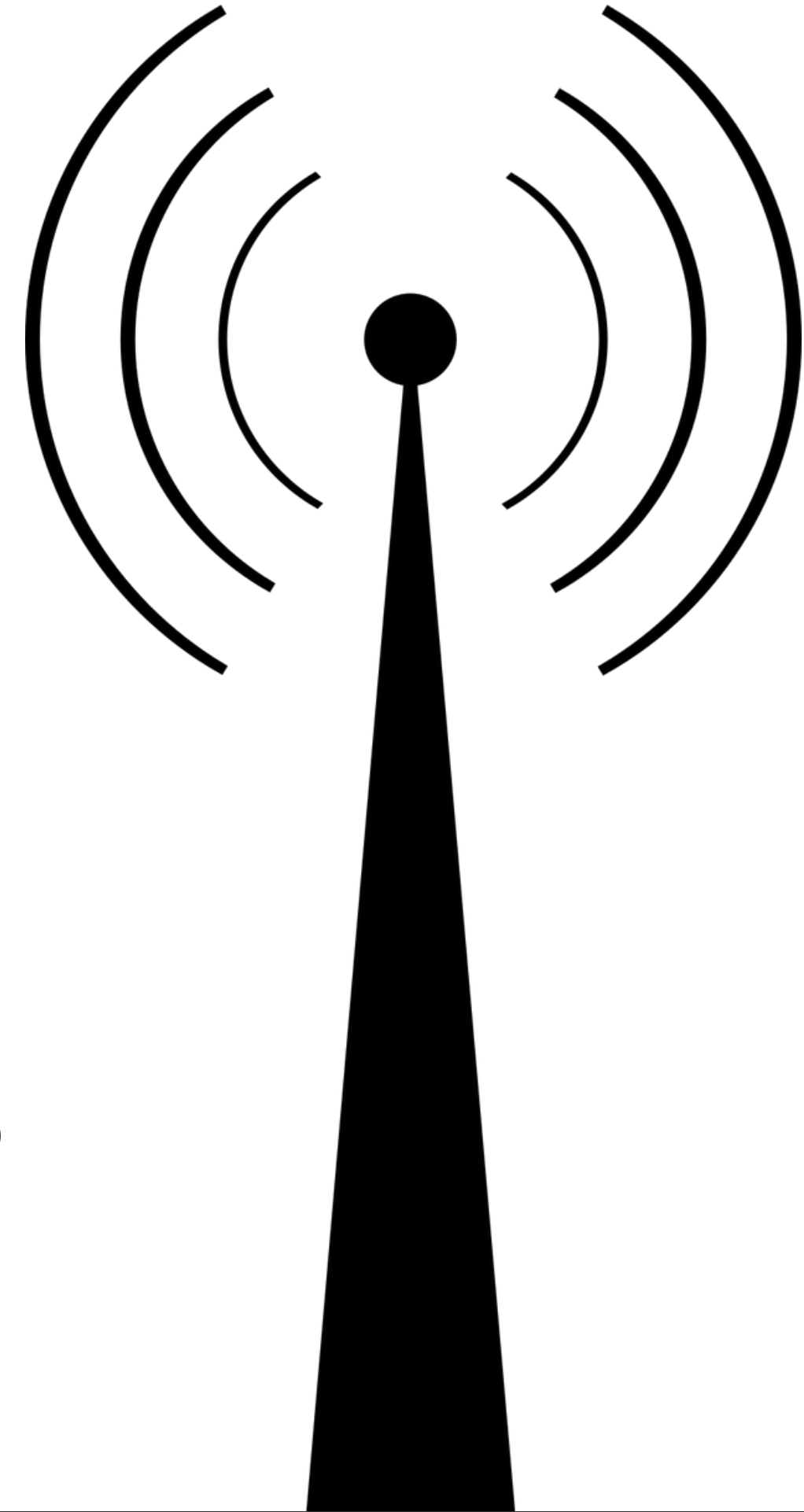
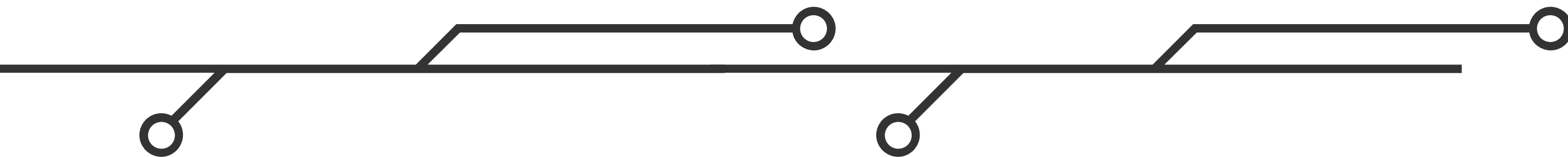


PHÂN ĐOẠN TIẾNG NÓI VÀ KHOẢNG LẶNG

WHAT IS A "SIGNAL"?

- A signal is a way to carry information, often using electricity, light, or radio waves.
- Imagine talking on the phone: your voice is turned into an electrical signal that travels through wires or air.
- There are two main types of signals: analog and digital.



ANALOG SIGNALS:

The Wavy World

- Analog signals are continuous, like a smooth, unbroken wave.
- They directly represent the information they carry.
- Think of a dimmer switch for a light: you can set it to any brightness level, smoothly.
- How they encode: The strength or frequency of the wave changes to match the information (e.g., a louder sound makes a bigger wave).

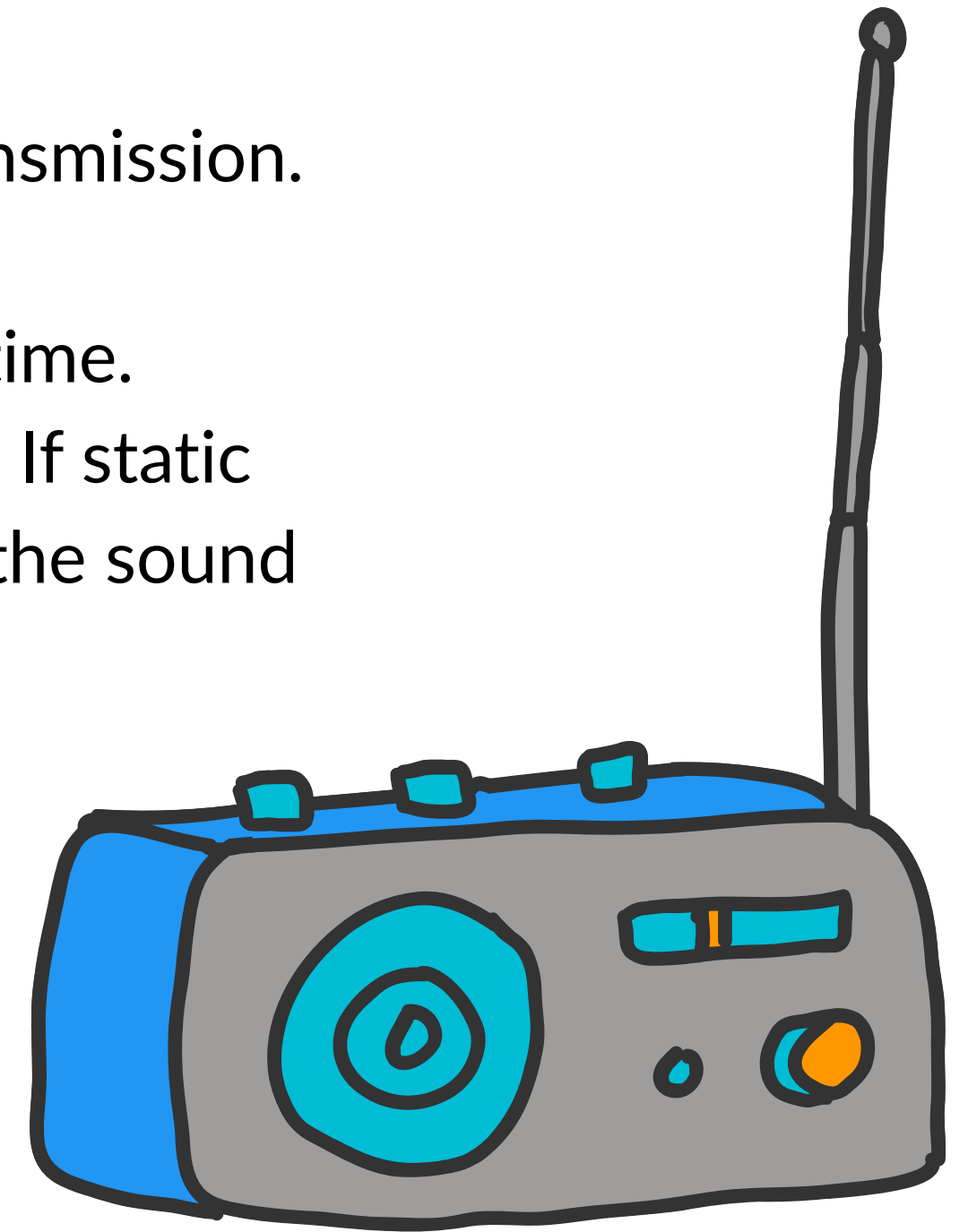
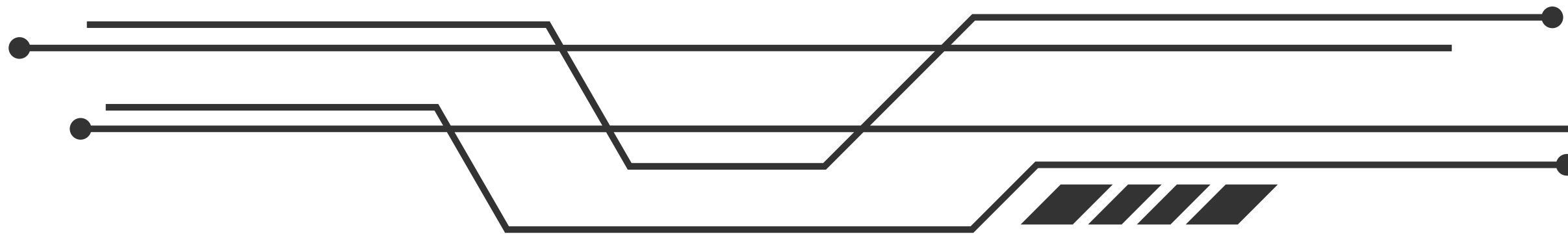
DIGITAL SIGNALS:

- Digital signals are made up of discrete, separate pieces of information, usually represented as binary code (0s and 1s).
- Think of a light switch: it's either ON (1) or OFF (0).
No in-between.
- How they encode: Information (like a sound or image) is converted into a sequence of 0s and 1s. Each 0 or 1 is a "bit."



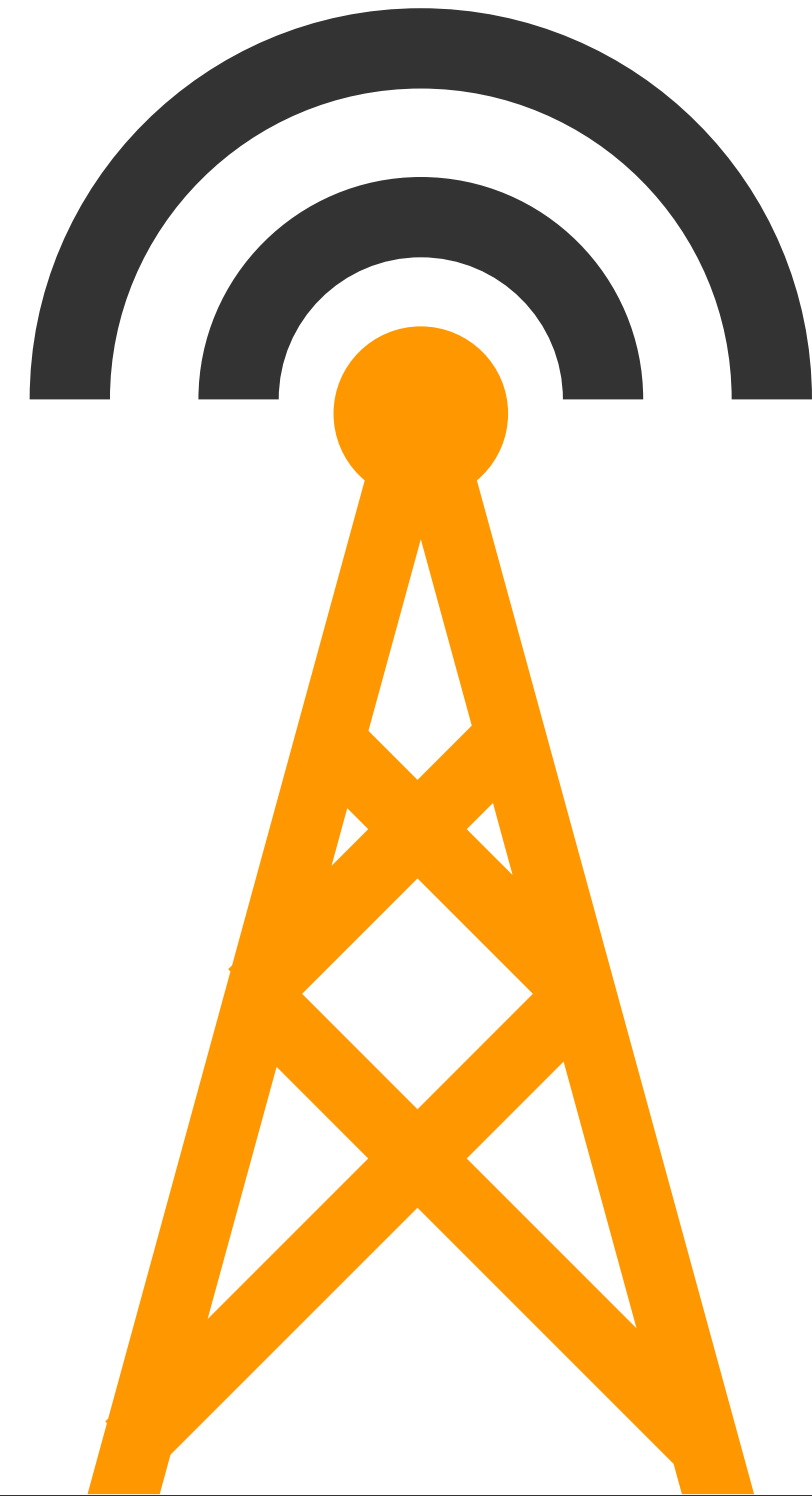
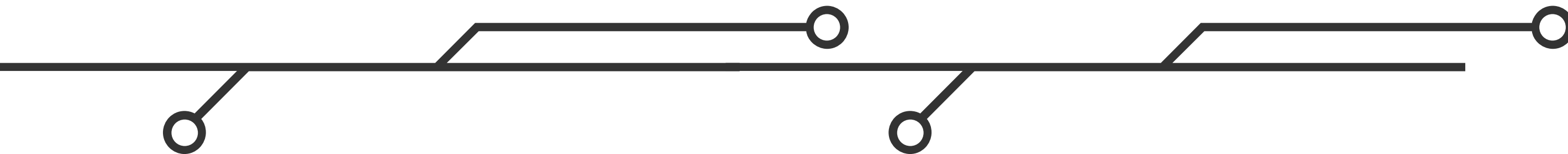
ANALOG'S CHALLENGE:

- **Noise:** Unwanted interference that gets added to a signal during transmission. Think of static on a radio.
- **Degradation:** When a signal loses quality or clarity over distance or time.
- **Problem:** With analog signals, noise and degradation are permanent. If static gets into a radio signal, it's there forever. If a cassette tape gets old, the sound quality gets worse.



WHY DIGITAL IS MORE RELIABLE

- Key Advantage: Digital signals can be regenerated.
- When a digital signal travels, it might pick up some noise. But because it's just 0s and 1s, a receiver can easily tell if a signal is a "strong 1" or a "weak 0."
- It can then recreate a perfect, clean version of the original 0 or 1.
- This means noise can be filtered out, and the signal can travel long distances without losing quality.



REAL-WORLD COMPARISONS

Music:



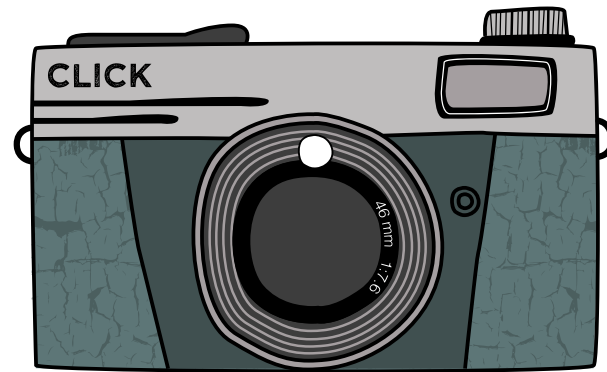
Analog: Cassette tapes, vinyl records (can get scratchy, hiss, wear out).



Digital: CDs, MP3s, streaming (clear sound, no static, can be copied perfectly).

REAL-WORLD COMPARISONS

Photography:



Analog: Film cameras
(photos can degrade, get
scratches).



Digital: Digital cameras, phone
cameras (clear, can be copied
endlessly without loss).

REAL-WORLD COMPARISONS

Communication:



Analog: Old landline phones (static, fading calls).



Digital: Cell phones, video calls (clearer calls, less interference).

